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18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

$H_0: p_1 = p_2 = p_3$, $H_a: p_1, p_2, p_3$ 不完全相等

$$E = \frac{200}{3} = 66.67$$

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(83 - 66.67)^2}{66.67} + \frac{(61 - 66.67)^2}{66.67} + \frac{(56 - 66.67)^2}{66.67} \approx 6.19$$

$$6.19 > \chi^2_{2, 0.05} = 5.991 \Rightarrow \text{Reject } H_0$$

⇒ 數據顯示人們對這三個入口的偏好有顯著差異 #

$$95\% \text{ C.I.} = \hat{p} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$= \frac{83}{200} \pm 1.96 \sqrt{\frac{0.415(1-0.415)}{200}}$$

$$= [0.3467, 0.4833]$$

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22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.¹ In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$H_0: p_1 = p_2 = \dots = p_1$, $H_a: p_1, p_2, \dots, p_1$ 不完全相等

$$E = \frac{200}{7} = 28.57$$

$$\chi^2 = \frac{(24 - 28.57)^2}{28.57} + \frac{(36 - 28.57)^2}{28.57} + \frac{(21 - 28.57)^2}{28.57} + \frac{(26 - 28.57)^2}{28.57} + \frac{(32 - 28.57)^2}{28.57} + \frac{(26 - 28.57)^2}{28.57} + \frac{(29 - 28.57)^2}{28.57} \approx 3.63$$

$$3.63 < \chi^2_{6, 0.05} = 12.596 \Rightarrow \text{Non-Reject } H_0$$

$\Rightarrow$ 我們沒有足夠證據支持心臟病發作機率會受到星期影響

18. Fatal Accidents Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large
Fatal	67	26	16
Not Fatal	128	63	46

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

H_0 : 致命事故的發生頻率 $\perp$ 汽車大小

H_a : 致命事故的發生頻率與汽車大小不獨立

$$E_{F,S} = \frac{109 \times 195}{346}, E_{F,M} = \frac{109 \times 89}{346}, E_{NF,S} = \frac{237 \times 195}{346}, E_{NF,M} = \frac{237 \times 89}{346}$$

$$E_{F,L} = \frac{109 \times 62}{346}, E_{NF,L} = \frac{237 \times 62}{346}$$

$$\chi^2 = \frac{(67 - 61.43)^2}{61.43} + \frac{(26 - 28.04)^2}{28.04} + \frac{(16 - 19.53)^2}{19.53} + \frac{(128 - 133.57)^2}{133.57} + \frac{(63 - 60.96)^2}{60.96} + \frac{(46 - 42.47)^2}{42.47}$$

$$= 1.884 > \chi^2_{2, 0.05} = 5.991 \Rightarrow \text{Non-Reject } H_0$$

$\Rightarrow$ 我們沒有足夠證據說明汽車大小與事故是否致命有關聯性

23. Telecommuting As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

Salary	Workdays at Home per Week		
	Less Than One	At Least One, but Not All	All at Home
Under \$50,000	38	16	14
\$50,000-\$74,999	54	26	12
\$75,000-\$99,999	35	22	9
Above \$100,000	33	29	12

- a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.
- b. Use Table 5 in Appendix I to approximate the p -value for this test of hypothesis. Does the p -value confirm your conclusions from part a?

a. H_0 : 薪資 $\perp$ 在家工作天數

H_a : 薪資 $\neq$ 在家工作天數

$$E_{11} = \frac{68 \times 160}{300}, E_{12} = \frac{68 \times 93}{300}, E_{13} = \frac{68 \times 41}{300}, E_{21} = \frac{92 \times 160}{300}$$

$$E_{22} = \frac{92 \times 93}{300}, E_{23} = \frac{92 \times 41}{300}, E_{31} = \frac{66 \times 160}{300}, E_{32} = \frac{66 \times 93}{300}$$

$$E_{33} = \frac{66 \times 41}{300}, E_{41} = \frac{14 \times 160}{300}, E_{42} = \frac{14 \times 93}{300}, E_{43} = \frac{14 \times 41}{300}$$

$$\begin{aligned} \chi^2 = & \frac{(38-36.21)^2}{36.21} + \frac{(16-21.08)^2}{21.08} + \frac{(14-10.65)^2}{10.65} + \frac{(54-49.01)^2}{49.01} + \frac{(26-28.52)^2}{28.52} \\ & + \frac{(12-14.41)^2}{14.41} + \frac{(35-35.20)^2}{35.20} + \frac{(12-20.46)^2}{20.46} + \frac{(9-10.34)^2}{10.34} + \frac{(37-39.41)^2}{39.41} \\ & + \frac{(29-22.94)^2}{22.94} + \frac{(12-11.59)^2}{11.59} \approx 6.446 < \chi^2_{6, 0.05} = 12.59 \end{aligned}$$

$\Rightarrow$ Non-Reject $H_0 \Rightarrow$ 我們沒有足夠證據說明薪資與在家工作天數有關

(同 (a))

$$b. p\text{-value} = P(\chi^2 > 6.45) > P(\chi^2 > 10.645) = 0.1 \Rightarrow \text{Non-Reject } H_0$$

⇒薪資水準 ↑ 在家工作天數